

Multi-Label Syntax

Fire multiple signals in a single transition:

```
transitions {
  // Write: 3 signals at once
  transition Idle -> WritePhase
    on label addr_valid,
    label data_valid,
    label ctrl_write;

  // Read: 2 signals at once
  transition Idle -> ReadPhase
    on label addr_valid,
    label ctrl_read;

  // Single-label (still valid)
  transition WritePhase -> Ack
    on label ack;
}
```

Comma-separate labels after on. Order does not matter.

When to Use Multi-Label

Multi-label transitions model scenarios where several signals assert *simultaneously* in one clock cycle:

- Hardware buses (AXI, SPI, PCIe)
- Grouped control signals
- Atomic compound actions

If events happen *sequentially*, use separate single-label transitions instead.

Single vs. Multi-Label

	Single	Multi
Signals/step	1	≥ 1
Syntax	on label a;	on label a, label b;
Use case	Sequential protocols	Parallel buses
Composition	Label syncs individually	All labels in set must sync

Example: AXI-Like Bus

Model a bus where address, data, and control assert together:

```
// AXI write = addr + data + write ctrl
transition Idle -> WritePhase
  on label addr_valid,
  label data_valid,
  label ctrl_write;

// AXI read = addr + read ctrl
transition Idle -> ReadPhase
  on label addr_valid,
  label ctrl_read;
```

The label `addr_valid` appears in both transitions but with different companions — this models two distinct bus operations sharing an address phase.

Other Transition Types

Besides labeled transitions, CTXDSL supports:

Epsilon transitions — silent/internal steps:

```
transition A -> B on epsilon;
```

Guarded transitions — conditions + variable effects:

```
transition A -> B
  guard (x > 0)
  effects { x = x - 1; };
```

Guards are covered in later tutorials (state variables & abstraction).

Composition with Multi-Labels

In *synchronous* composition, a multi-label transition synchronizes on **all** of its labels that are shared with the other automaton. Non-shared labels in the set combine freely.

Rule of thumb: if *any* label in the set is shared, the partner must be able to participate on that label in the same step.

Formulas on Multi-Label Systems

Labeled modalities work with individual labels:

```
// Does an addr_valid transition exist?
< labels = { addr_valid } > true

// All ack transitions end safely
[ labels = { ack } ] Safe
```

A multi-label transition **on a, b** is matched by modalities filtering on **a or b**.